

Marty's project Curriculum

process-related competencies

content-related competencies

Prior knowledge to confirm?

- What is a motor
- What is a sensor
- Notions around batteries?
- Electricity and circuits already done?
- Auftrag 2.1: Intelligenz?
- Auftrag 2.2: Charakter-Gespräch
- Auftrag 2.3: Soekia
- Auftrag 2.4: Bot or not?
- Auftrag 2.5&2.6: Prompt Engineering 1&2
- Auftrag 2.7: TeachableMachine
- Auftrag 2.8: Machine Learning Teil 1
- Auftrag 2.9: Machine Learning Teil 2 (python, dataframes, panda)
- Auftrag 2.10: Machine Learning 3 (histograms)
- Auftrag 2.11: Machine Learning Teil 4 (model)
- Auftrag 2.12: Machine Learning Teil 5 (model evaluation)
- Auftrag 2.13: Projektarbeit Künstliche Intelligenz
- Auftrag 2.14: System Prompts
- [Auftrag 2.16: Build-a-Bot](#)
- What experience with programming?
 - What is a loop
 - What is a variable
 - What is a function
- Jupyter Notebook in Teams? Know python?

Learning Objectives NIT 8 (NwT) – M/R/E

- Fundamentals of programming and wiring a microcontroller
- Hazards to eyes and ears
- Possibly 3D printing and creating a 3D model for individual parts (E-standard)
- Introduction to the Arduino microcontroller
- Introduction to a programming environment

- Control systems using microcontrollers
- Handling woodworking tools
- Building a wooden enclosure

Outlook:

- The role of microcontrollers in everyday life (1 lesson)

Skill Development:

- Technical drawing (4 lessons)
- Woodworking (6 lessons)
- Soldering (1 lesson)
- Introduction to electronic circuits and digital functions of the microcontroller (8 lessons)
- Reading datasheets
- Optional: Working with a CNC milling machine

Project Phase:

- Planning and creating an attractive "disco room that rocks" in groups (4 lessons)

Curriculum standards

- [3.1.2 Algorithms](#)

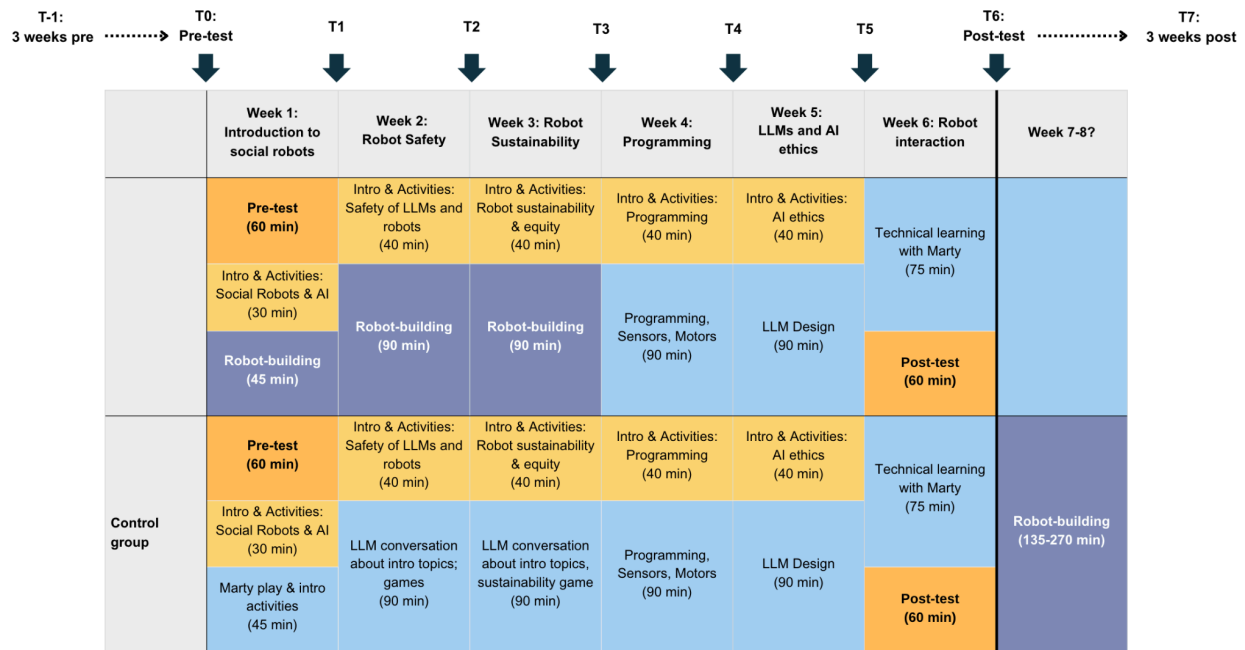
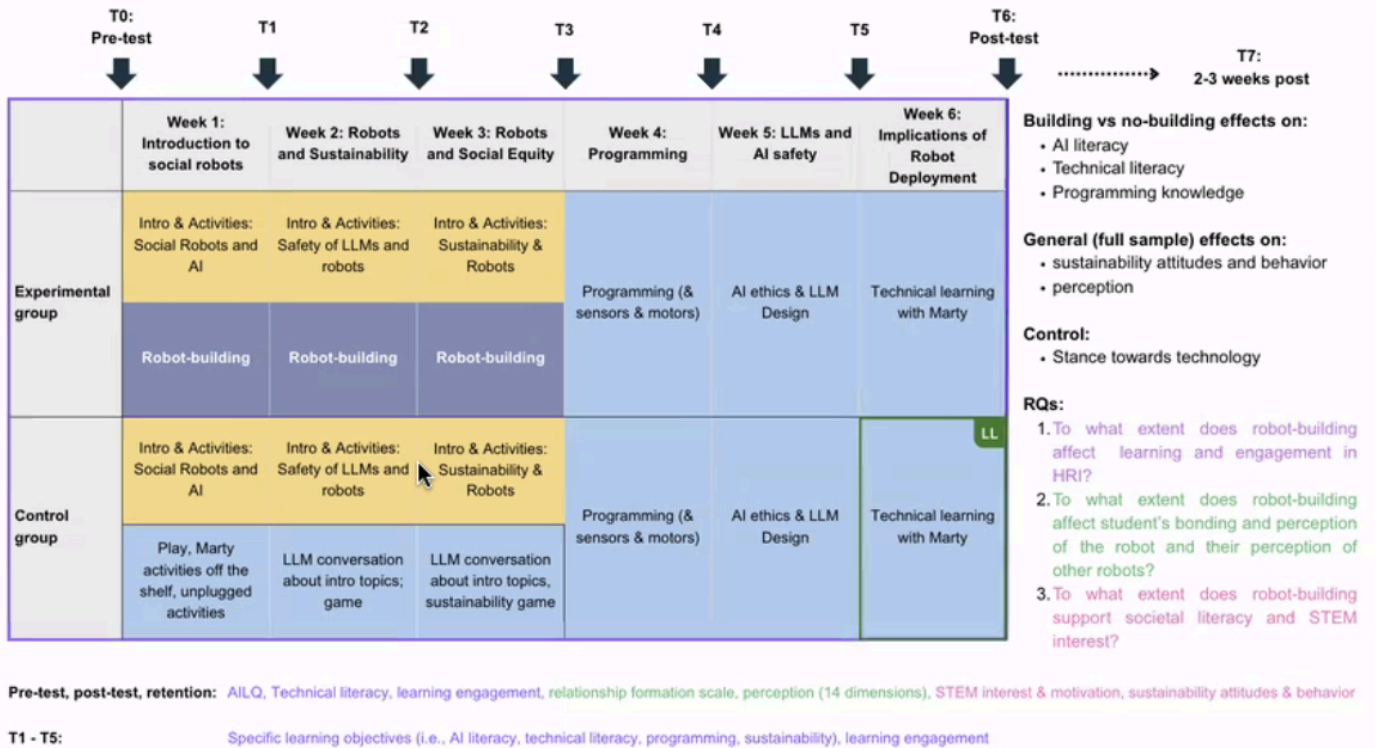
- (1) Use logical links (AND, OR, NOT) in conditions of loops and branches
- (2) Use random numbers in your own programs (e.g. to simulate dice results or to make a game more varied)
- (3) use an index-based data structure for storing and processing similar data (also by iteration)
- (4) basic algorithms on an index-based data structure (e.g. filling with values, maximum search, summing)
- (5) Use subroutines to structure program code and avoid redundant code

- [3.1.4 Information society and data security](#)

- **(1) explain a currently used technology with which personal data is collected (e.g. web tracking, cookies, geodata)**
- (2) Describe ways to restrict the collection of personal data (e.g. anonymous browsing, app rights management, location sharing)

- [3.2.1 Ways of Thinking and Working in Science and Technology: Systems and Processes](#)
 - (1) Analyze systems and describe them through system boundaries and subsystems (e.g. living beings, machines, solar system)
 - (2) Describe energy, material and information flows between subsystems (e.g. metabolism)
 - (3) Subsystems with regard to their external functions (black box thinking); e.g. sensory cell, battery)
- [3.2.2 Systems and Processes](#)
 - **(1) Name the essential assemblies of a machine and explain the interaction (e.g. drill/driver, hand mixer)**
 - (4) a technical system under socio-technical aspects (e.g. standardization and safety) and human-social aspects (e.g. aesthetics, ergonomics and economic efficiency)
 - (5) Create circuit diagrams with standard-oriented symbols manually or digitally
 - (12) Realize tasks with the logic functions AND/OR with switches
 - **(15) detect physical quantities with sensors (e.a. micro-tass and LDR)**
 - **(16) Realize actuator controls (e.a. LED and motor)**
 - **(17) Realize controls with sensors and actuators (e.g. light and dark circuit, alarm system, traffic light switch)**
 - **(18) Describe simple controls and controls with block diagrams**
- [3.2.2.2 Movement and locomotion](#)
 - (5) construct and manufacture an object with drive
- [3.2.3.3 Product development](#)
 - (1) develop, construct and present a product with a defined function and a specific property in a norm-oriented manner (e.g. wind turbine, machine)
 - (2) Explain analogies between technical products and natural systems (e.g. lotus effect, thermal insulation, stability of constructions)
 - (3) Select and use raw materials in a resource-saving manner (cut, life cycle assessment)
 - (4) to manufacture a product with tools and machines (process for separating, joining, forming, e.g. computer-aided manufacturing)
 - (5) Evaluate the function and characteristics of a product
- Physics: [3.2.3 Energy](#)
 - (1) Describe the basic properties of energy (e.a. Energy conservation)
 - (2) Give examples of energy transmission chains in everyday life and technology and describe them qualitatively (including mechanical, electrical or thermal energy transmission)
 - (7) Describe the relationship between energy and performance ($P=E/T$)
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- [3.2.4.1 Information reception by senses and sensors](#)
 - (1) describe the possible uses of sensors (e.g. blood pressure monitor, hygrometer, anemometer)
 - (2) Compare the design and functioning of a sensory organ with a corresponding technical sensor (e.g. eye with digital camera, ear with microphone)
 - (3) describe the danger to the eye or ear due to overload and explain health limits
- [3.2.4.2 Collection and evaluation of data](#)
 - (3) represent an optical or acoustic spectrum (e.g. solar spectrum)
- [3.2.4.3 Information processing](#)
 - (2) represent and describe the principle of control (e.g. Robotics)
 - (3) Describe elements of a programming language (e.g. condition, branching, loop, time arm)
 - (4) Represent algorithms for time- and sensor-controlled processes in a programming language and thus implement control sequences (e.g. traffic light control, robotics)
- Ethics:
 - [3.1.3.1 Acting in the media-mediated world](#)
 - (1) examine and evaluate the influence of the media and media networking on their lives and those of others (e.g. B. everyday life, social relations)
 - (2) Work out and discuss opportunities and risks of media use in their living environment (e.g. learning opportunities, information growth, communication, entertainment, cyberbullying, addiction, privacy)
 - (3) Identify and evaluate the interests and needs of participants and those affected in the use of media in exemplary situations (e.g. in relation to privacy, data protection, information, entertainment, respect)
 - (4) explain the prescribed legal basis for dealing with media and use examples to show and justify the responsibility of the individual and of groups in the use of media (e.g. freedom of expression and of the press, fundamental rights, data protection, youth protection law)
 - (5) Represent and evaluate options for a responsible and self-determined use of media (e.g. in relation to data protection, use, disposal)
 - [3.1.5.2 Man, Nature, Technology](#)
 - (3) present and discuss various effects of technology on people and nature and the consequences of increasing technification with regard to possible conflicts of values (e.g. self-determination, comfort, sustainability)
 - (4) Examine aspects of freedom and responsibility in human dealings with nature and technology using various examples, develop possibilities for a responsible and sustainable use of technology and nature (e.g. protection of biodiversity, conservation of resources) and argue for these with values and norms



Planned assessment

Pre/Post	Weekly
AI/Technical Literacy	Enjoyment / Ease of learning
Sustainability attitudes and behavior	Learning objectives
Robot perception (competence, intelligence, friendliness, socialness, warmth)	Ease of teaching the class
Relationship formation with robot	
Attitudes towards robotics and learning AI / value of AI literacy	
STEM interest / sense of belonging / ability beliefs	
Learning engagement / attention	

Detailed plan

[Week 1: Constructing Marty - Getting Started / Introduction to social robots \(2h05\)](#)

[Introduction: 15 minutes](#)

- Warm up: robots classification cards
- Presenting the project:
 - Definition of a social robot and how it is different from other robots and virtual agents
 - Projects goal and timeline
- Who is Marty
- What it can do
- What we will achieve

Pre-Test: 60 minutes

- Perception of robot (i.e., competence, socialness, friendliness)
- Knowledge of sustainability, social equity, AI

Building Session: 45 min

- Legs

Assessment and reflection (5 min)

- Assessment documents

- Reflection guide

Week 2: Safety of LLMs and Robots (2h15)

Teacher's material:

- [Artificial Intelligence](#)
- In english : <https://code.org/en-US/artificial-intelligence>

Introduction: 10 minutes

- Intro to Large Language Models (LLMs)
 - Understand the basic AI concepts and their origins such as machine learning, deep learning and neural network. (optional: short history of AI and LLMs)
 - Differentiate LLMs from other AIs
 - LLMs are trained on very large amounts of text-based data, not on sensors data (those are other models often called embodied or robotics foundation models)
 - It appears that LLMs nonetheless create representations and can reason
 - LLMs are very good at processing large amounts of text data and present them in a digestible and understandable way
 - LLMs can identify patterns and classify information that humans cannot, for example identifying humans' mood and emotions
- Limitations of LLMs
 - Hallucinations
 - Biases
- Uses of LLMs (with concrete examples)
 - Brainstorming and Co-creation
 - Searching, summarising and detecting patterns
 - Tutoring, coaching and personalization of learning
 - Evaluations, screening and gradings
 - AI Agents
 - Other

Activity: 30 minutes

- <https://teachablemachine.withgoogle.com/> (to confirm)

Experimental group: Building Session (Hardware): 90 minutes

- Continue legs
- Arms

Control Group:

- Work with this? <https://studio.code.org/courses/oceans/units/1/lessons/1/levels/2>
 - Slides: <https://appcamps.slides.com/appcamps/kimale>

Assessment and reflection (5 min)

- Assessment documents
- Reflection guide

Week 3: Sustainability and Social Equity of Social Robots (2h15)

Introduction: 10 minutes (probably not all this)

- Intro
 - A robot requires electronics, often a battery, building material which could be anything: plastic, wood, silicone, metal, etc.
 - Sustainability is when we can keep doing what we are doing without longterm negative effects (planetary boundaries?)
- Plastic
 - Plastic is a magical material that is very light, very cheap, can come in many shapes, and can be durable. It is so successful that we produce so much plastic (give numbers) that we can't manage it.
 - The problem with plastic is not CO2 (give numbers and compare with other materials), but pollution, which impacts health and biodiversity. There will be no problem with plastic if quantities were little and we could recycle it all. However, recycling also has a cost, energetically, ecologically and economically.
- Electronics : silicon, metal, plastic
 - It is very difficult and expensive to recycle that. We currently don't have a good solution at scale. Product design can help a lot to increase recyclability, for example by making the materials easily separable and repairable
- Batterie
 - Lithium is difficult to get (show world production) and creates geopolitical tensions
 - We are starting to see lithium recycling but it is very new
- Marty Robot by RoboBots
 - How Marty's parts are built and where
- Positive and negative feedback loops
 - Sustainability is closely related to ecological systems
 - Ecological systems are governed by non linear relationships with positive (reinforcing) loops and negative (stabilising) loops
 - These feedback loops have tipping points, which are irreversible (at human scale) changes to the system reaching a new equilibrium.
 - Plastic pollution is likely gone beyond tipping points
- Equity and Society
 - Equity focuses on results, while equality on means
 - AI and robotics will substantially change societies, sometimes positively, sometimes negatively
 - Costs, knowledge, skills and infrastructure are barriers to equitable benefits from AI

- Cultural differences and unequal representation in training data impact quality of AI products used by non western countries
- In summary
 - The problem is related to the scale of production and how we design products – whether it includes the whole lifecycle of the product or not.
 - The pace of innovation and economical growth related pressures shorten lifecycle of products, increasing the level of production and waste management.
 - Recycling has a cost and an impact and biodegradability is often second to reduced consumption.
 - Optional: Kaya identity (very good at explaining : Expanding the scope of sustainability: Broaden sustainability education beyond a sole focus on and waste management to include critical concepts like mindful and reduced consumption)

Activity (30 min):

- Create a new game, for example "[Robot Crisis Supply Chain Game](#)" combining the Beer Game mechanics with sustainability pressures:
 - 4 Player Roles: Lithium Miner, Electronics Manufacturer, Robot Company, School District
 - Random Events: Climate protests, mining strikes, new recycling tech, regulation changes
 - Hidden Information: Each player only sees their piece of the puzzle
 - Victory Condition: Keep robots flowing to schools while meeting sustainability targets

Experimental group: Building Session (Hardware): 90 minutes

- Finish assembly

Control Group:

- Climate fresque game?

Assessment and reflection (5 min)

- Assessment documents
- Reflection guide

[Week 4: Interaction Through Programming \(2h15\)](#)

[Marty's guide for teachers](#)

- [Intro to python](#)
- [Marty says cards](#)

Intro (10 min)

- Basic concepts review:
 - Functions, variables, parameters and commands
 - Libraries
 - Loops, conditions (if.. then)
 - Data structures
- Architecture
 - Marty's architecture
- Motors
 - a motor transforms electric energy in movement using electromagnetic field
- Sensors
 - a sensor captures information about the environment and transmits it
 - there are many types of sensors
 - Marty has 2 sensors (infrared and color sensor)
 - an infrared sensor reacts at temperature (reminder about electro-magnetic waves?)
 - color sensor reacts for colors (sends a blue signal when sees blue, etc.
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First programming activities (30 min)

- Exploring Marty
- Make Marty move and dance using bottoms, then sequence, then coding blocks, then C++
- Not GOING OVER THE EDGE (robotical lesson)
- Other activities by robotical

Robot programming (90 min)

- Programming emotions
 - Make Marty move to look happy, sad, worried and angry
- Programming dance moves and a choreography
 - Make Marty move with music
 - Coordinate with other groups to create a choreography of the 10 Martys. Test it and when ready record a video of the performance.

Assessment and reflection (5 min)

- Assessment documents
- Reflection guide

Week 5: AI Ethics (2h15)

Intro (10 min)

- Ethics
 - Definition of ethics: Ethics is the branch of philosophy that deals with what is morally right and wrong, and how decisions affect people and society.

- There is intentional and unintentional harm; physical and psychological harm (examples with non-social robots)
- Accessibility and sustainability concerns are part of ethics
- Ethical technology ideally includes considerations of human development
- AI Ethics
 - Transparency and explainability
 - Data privacy
 - Alignment (how LLMs could be moral and ethical) & Mental Health
 - Over-Reliance on AI (long term cognitive, socio-emotional, physical and behavioral impacts)
 - Manipulation and Persuasion (emotions, feelings and desires)
 - Misinformation and Content Accuracy
 - Sustainability impacts (energy, materials and water)
 - Equity and accessibility
 - Legal responsibility & intellectual property (optional)
 - Other social impacts (job market, cultural norms, etc.) (optional)
- Other LLMs best practices
 - Prompt Engineering
 - Co-thinking
 - Reminder: LLMs can't be reliable sources of information → always find and check the sources

Discussion/Activities about safety of AI (35 min)

- To define, Or more time on next activity

LLM personalization, safe-guarding and testing (55 min)

- Testing existing safeguards
 - The LLMs have safeguards (and students test if the safeguards work. Scenarios will be provided and students might create new ones. They would compare responses from commercial LLMs and the one provided and identify the differences)
- Personalization
 - Students will create a personalized scenario of interaction (ex. cooking master, dj, etc.) and will define a “contract”/ a “frame” with the LLMs about how it would behave in certain situations.
- Creating new safeguards
 - Students will design how the LLMs should behave in a certain situation (ex. a user insults it, or it hallucinates, or there is a bias) and created a procedure to integrate the safeguard and then test it.

Assessment and reflection (5 min)

- Assessment documents
- Reflection guide

Getting to Know Your Marty

- Ask about sustainability, social equity, AI safety/data

Week 6: Using Classroom Design to Discover Responsible Robot Deployment

Intro (10 min)

Learning with Marty about relevant topic (e.g., microcontrollers) (45 min)

Post-Test: 60 min

- Perception of robot (i.e., competence, socialness, friendliness)
- Knowledge of sustainability, social equity, AI

Classroom discussion and reflection: 20 min